REGULARIZED SPECTRAL STATISTICS FOR PRIME RACES AND LOW-ZERO TRANSIENT REVERSALS

SHIN-YA KOYAMA AND SAAR SHAI

Abstract. We study a proposed Gaussian-regularized statistic for prime races and a separate finite-scale reconstruction of ordinary prime-count races. Character orthogonality gives the exact selector

$$\frac{1}{\varphi(N)}\sum_{\chi}(1-\overline{\chi(a)})\chi(x) = \mathbf{1}_{x\equiv 1 \pmod N} - \mathbf{1}_{x\equiv a \pmod N}.$$

For the regularized statistic we formulate an explicit two-parameter analytic target with $T = T(x)$; the fixed-T asymptotic is not asserted, because the diagonal coefficient depends on T and a summed off-diagonal estimate remains to be proved. The proposed extremality of $-1 \pmod N$ and recessiveness of $1 \pmod N$ are therefore stated as conjectural fine-structure principles.

Independently, we analyze ordinary prime-count data for $N \in \{7, 8, 11, 19, 23\}$ at 438 checkpoints through 3×10^{14}. The extended computation agrees with an independently checked baseline in 567 of 567 shared cells through 1.3×10^{13}. Low-zero explicit-formula truncations using 25 positive zeros per nonprincipal character correlate with the observed top-decade -1 trajectories at levels 0.826–0.971. Nevertheless, frequent rank and leader changes persist. In particular, an Arb sign-change computation certifies a critical-line zero near 0.018956399080226143 for the odd character of Conrey index 13 modulo 19. These results explain the observed transients but do not establish an eventual ordering of ordinary prime counts or validate the proposed regularized asymptotic.

1. Introduction and Main Results

Since Chebyshev's seminal observation in 1853 that primes congruent to $3 \pmod 4$ appear more frequently than those congruent to $1 \pmod 4$, the phenomenon of prime number races and residue bias has been a central topic in analytic number theory. Rubinstein and Sarnak [4] established a rigorous probabilistic framework for prime biases under the Generalized Riemann Hypothesis (GRH) and the Grand Simplicity Hypothesis (GSH). Later, Feuerverger and Martin [2] investigated weighted prime sums and observed subtle deterministic hierarchies governed by special values of Dirichlet L-functions.

1.1. Beyond Chebyshev's Bias: Discovering the Fine-Structure Hierarchy.

Classical studies on prime number races, stemming from Chebyshev's original observation, predominantly attribute the prime bias to the accumulation of prime squares ($k = 2$), which creates a systematic deficit in quadratic residue classes (e.g., $3 \pmod 4 > 1 \pmod 4$). However, this classical mechanism suffers from a fundamental limitation: *it fails to distinguish or predict any bias among residue classes that share identical quadratic character values.*

For instance, modulo $N = 8$, every non-trivial residue class satisfies $x^2 \equiv 1 \pmod 8$. Consequently, the traditional prime-square effect predicts complete equiprobability among the non-principal classes $3, 5, 7 \pmod 8$.

Date: August 16, 2026.

The regularized construction below should be read as an analytic program, not as a completed asymptotic theorem. Its arithmetic selector is exact, but the Gaussian argument must control two coupled parameters and the total off-diagonal contribution. We therefore separate three levels of evidence: proved finite character algebra, a conjectural regularized limit, and verified finite computations for ordinary prime counts.

1.2. Motivation and relation to Aoki–Koyama.

The use of the Deep Riemann Hypothesis (DRH) in prime-bias problems is motivated by Aoki and Koyama [1]. Here it motivates a proposed spectral regularization across Dirichlet characters. No transfer from this regularized statistic to eventual ordinary-count dominance is assumed.

1.3. Gaussian localization and its two parameters.

For a scale $T > 0$ and a prime power p^k, consider

$$h_{p^k,T}(\gamma) := \frac{\cos(k\gamma \log p)}{p^{k/2}} e^{-(\gamma/T)^2}. \tag{1.1}$$

For each fixed p^k and T this is a Schwartz function. The difficulty is uniformity after summing over all prime-power pairs up to x: logarithmic separations between distinct prime powers shrink as x grows. Accordingly, x and T must be treated jointly.

1.4. Corrected selector and analytic target.

Let $G = (\mathbb{Z}/N\mathbb{Z})^\times$ and let G^* be its complete character group.

Lemma 1.1 (Character selector)**.** *For $a, x \in G$,*

$$\kappa_a(x) := \frac{1}{\varphi(N)} \sum_{\chi \in G^*} (1 - \overline{\chi(a)})\chi(x) = \mathbf{1}_{x=1} - \mathbf{1}_{x=a}. \tag{1.2}$$

Proof. Character orthogonality evaluates the first term at $x = 1$. Since $\overline{\chi(a)} = \chi(a^{-1})$, the second evaluates at $a^{-1}x = 1$, equivalently $x = a$. Without the conjugate, the negative class would instead be a^{-1}. □

Choose logarithm branches in conjugate pairs and define, whenever those branches are specified consistently,

$$B_N(a) := \frac{1}{\varphi(N)} \sum_{\chi \neq \chi_0} (1 - \overline{\chi(a)}) \operatorname{Log}_\chi L(1, \chi), \tag{1.3}$$

with $\operatorname{Log}_{\bar\chi} L(1, \bar\chi) = \overline{\operatorname{Log}_\chi L(1, \chi)}$. This pairing makes $B_N(a)$ real. It also gives $B_N(1) = 0$; any proposed "nadir" statement for the identity class therefore needs a sign convention and a separate inequality, rather than following from the selector alone.

Conjecture 1.2 (Two-parameter regularized limit)**.** There is a precisely normalized regularized difference $R_{N,a}(x, T)$ and a growth regime $T = T(x)$ such that, as $x \to \infty$,

$$R_{N,a}(x, T(x)) = C_{N,T(x)} B_N(a) + \mathcal{E}_{N,a}(x, T(x)), \tag{1.4}$$

where $C_{N,T} > 0$ is explicit and real and $\max_{a \in G} |\mathcal{E}_{N,a}(x, T(x))| \to 0$. The regime must justify every order interchange and a summed off-diagonal bound. In particular, the factor $T/(4\sqrt{\pi})$ arising in the diagonal Gaussian calculation is retained in $C_{N,T}$ unless an actual cancellation is proved.

If Conjecture 1.2 is proved and the values $B_N(a)$ are distinct, their ordering would govern that regularized statistic only. An eventual ordering of ordinary counts would require a separate transfer theorem with explicit error control.

Conjecture 1.3 (Apex and nadir fine structure)**.** Within a fixed quadratic family, $-1 \pmod N$ is extremal for the consistently normalized values $B_N(a)$. With the same sign convention, the identity class $1 \pmod N$ is the opposite endpoint. Both assertions concern the proposed regularized hierarchy; neither is inferred from finite ordinary-count data.

1.5. **Primary and secondary scales.** The motivating two-tier template is

$$\psi(x; N, a) = \frac{x}{\varphi(N)} - \frac{N_Q(a)}{\varphi(N)}\sqrt{x} + C'_{N,T} B_N(a) + \text{remainder}, \tag{1.5}$$

where $N_Q(a)$ is the number of unit square roots of a modulo N. The $\sqrt{x}$ term is the primary Chebyshev-bias scale; the proposed bounded $B_N(a)$ term is secondary. Formula (1.5) is a structural target, not an asymptotic expansion established here: an error of order $\sqrt{x}/\log x$, for example, would be larger than the proposed secondary term and could not certify its ordering.

2. Gaussian regularization and finite-scale spectral diagnostics

2.1. **What remains to prove for the Gaussian statistic.** For a fixed distinct pair of prime powers, Fourier inversion supplies Gaussian decay in $T|\log(q^m/p^k)|$. It does not supply a positive separation constant uniform over all pairs below a growing bound x. A proof of Conjecture 1.2 must therefore give a complete summed off-diagonal estimate in its chosen $T(x)$ regime, justify the prime/zero/order interchanges, control the $k = 1, 2$ terms and the $k \geq 3$ tail, and include the Archimedean, conductor, and imprimitive Euler-factor contributions. Principal character cancellation from Lemma 1.1 is exact, but by itself it does not evaluate the remaining sums.

For the same reason, a finite-x plot of a proposed $R_{N,a}(x, T)$ against $C_{N,T} B_N(a)$ is not reported here. Such a plot becomes meaningful only after the statistic, branch convention, normalization, and $T(x)$ regime are fixed by the analytic statement; otherwise the constant can be rescaled after the fact.

2.2. **Ordinary-count data and reconciliation of the former Table 3.** The ordinary-count dataset contains 438 logarithmically spaced checkpoints for $N \in \{7, 8, 11, 19, 23\}$ through 3×10^{14}. A fresh comparison with the previous independently checked baseline gives 567 exact integer agreements in 567 shared cells through 1.3×10^{13}. The extension above that bound is one full run and is not described as an independent replication.

Table 1 replaces the former Table 3 directly from the raw class counts. It corrects eight cells at 1.3×10^{13}, corrects the sign of the $N = 19, a = 10$ entry at 1.3×10^{11}, and restores the omitted class $a = 20$ modulo 23. Each entry is $D_N(a) = \pi(x; N, a) - \pi(x; N, 1)$; the classes shown are all quadratic nonresidues.

At the final checkpoint, -1 leads only for $N = 7$ and 23. For $N = 7, 8, 11, 19, 23$, its respective ranks are

$$1/3, \qquad 3/3, \qquad 3/5, \qquad 6/9, \qquad 1/11.$$

This finite observation neither supports nor contradicts Conjecture 1.3, which concerns a separate regularized statistic.

2.3. **Low-zero reconstruction of the transient trajectories.** For a unit class a, define the normalized ordinary race

$$E_N(x;a,1) := \frac{\varphi(N)\log x}{\sqrt{x}}\big(\pi(x;N,a)-\pi(x;N,1)\big). \tag{2.1}$$

The finite GRH-form truncation used in the computation is

$$\widetilde{E}_{N,K}(x;a,1) := C_N(1)-C_N(a)-\sum_{\chi\neq\chi_0}\sum_{j=1}^{K} 2\,\mathrm{Re}\left(\frac{(\overline{\chi(a)}-1)e^{i\gamma_{\chi,j}\log x}}{\frac{1}{2}+i\gamma_{\chi,j}}\right), \tag{2.2}$$

where $C_N(a)$ counts unit square roots of a and K denotes the first K positive ordinates for each nonprincipal character.

The ordinates were generated through height 80 by two PARI/GP mesh runs (parameters 64 and 96) and required to agree character by character. Every used ordinate also satisfies a direct residual bound below 10^{-28}. These are robustness checks within one implementation, not a proof of zero completeness or GRH. Three first-zero anchors modulo 8 independently agree with earlier `mpmath` values to 10^{-12}.

Computational Result 2.1 (verified finite computation)**.** *The checked pipeline produces 13,578 reconstruction rows. On the 53-point window* $x \geq 3\times 10^{13}$*, the correlations between* $\widetilde{E}_{N,25}(x;-1,1)$ *and* $E_N(x;-1,1)$ *are*

$$0.965, \quad 0.931, \quad 0.939, \quad 0.971, \quad 0.826$$

for $N = 7, 8, 11, 19, 23$*. Across the 52 consecutive intervals, the observed rank of* -1 *changes* $17, 23, 30, 17, 39$ *times, respectively.*

2.4. **The very low odd-character mode modulo 19.** For the primitive character of Conrey index 13 modulo 19, of order 18,

$$\gamma = 0.0189563990802261\ldots, \qquad \chi(-1) = -1. \tag{2.3}$$

An independent Python FLINT/Arb computation gives opposite, sign-definite Hardy-Z endpoint balls across a bracket of width 5.94×10^{-84} and a midpoint L-value upper bound below 3.55×10^{-85}. This certifies existence of a critical-line zero in the bracket, but not uniqueness, completeness, or GRH. Its top-decade RMS contribution to the -1 race is 6.719, and its correlation with the centered observed curve is 0.728.

Using 100 positive ordinates for every nonprincipal character modulo 19 raises the -1 correlation from 0.9715 at $K = 25$ to 0.9925 at $K = 100$; the reconstruction still has 14 rank changes and 7 leader changes. Therefore the previous $\gamma \approx 1.74$ one-mode calculation and its $e^{33.4}$ "settling scale" cannot be retained. The data show a slow active mode and do not identify 3×10^{14} as a stabilization threshold.

The defensible finite-scale conclusion is that the ordinary races remain spectrally coherent but dynamically unsettled through 3×10^{14}. Low-zero truncations explain much of the observed motion; they do not prove an eventual ordinary-count ordering.

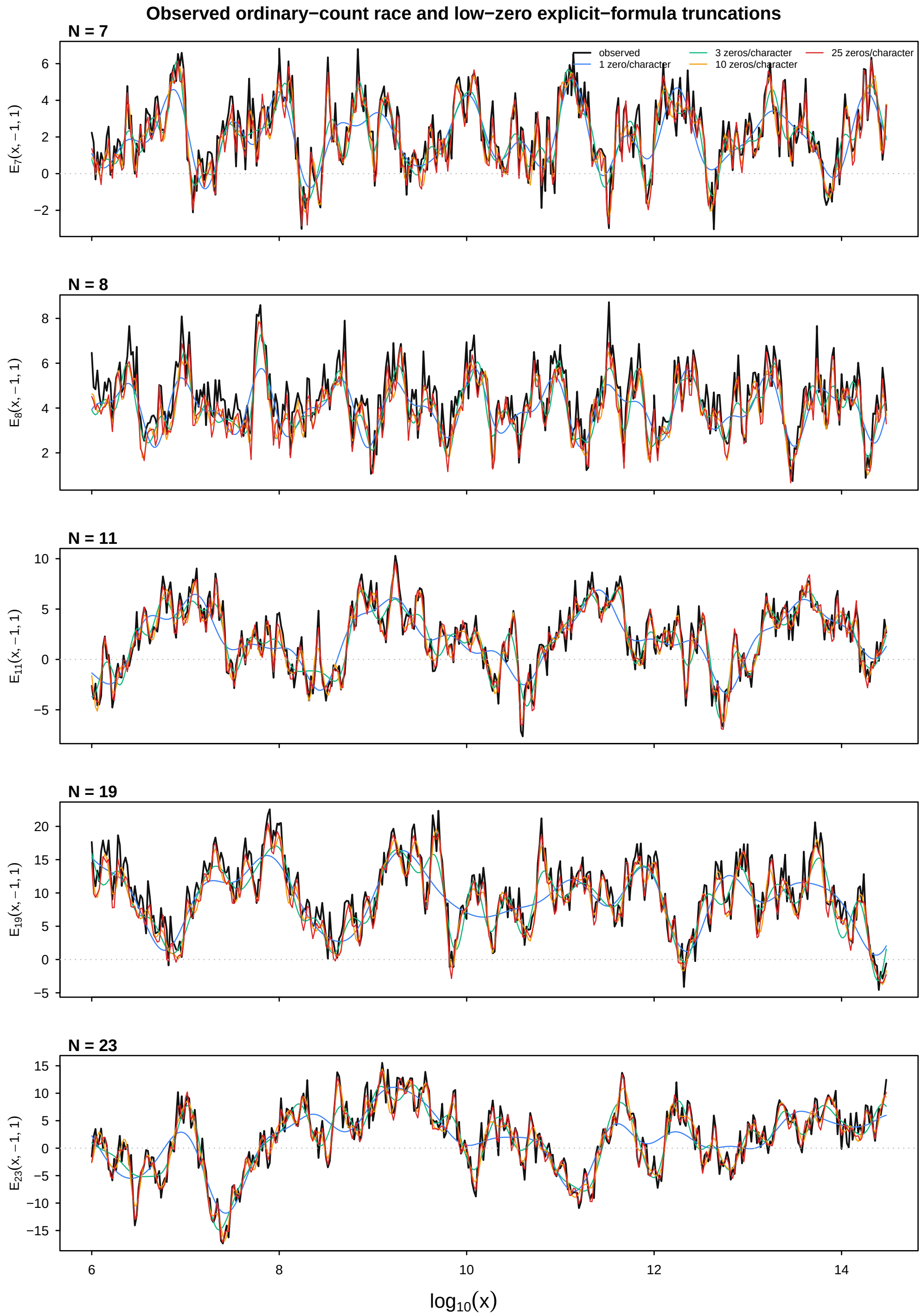

FIGURE 1. Observed normalized ordinary-count races (black) and explicit-formula truncations using $K = 1, 3, 10, 25$ positive zeros per nonprincipal character. The panels show the class $a = -1 \pmod{N}$.

Table 1. Reconciled ordinary prime-count differences from raw class counts.

N	x	$a : D_N(a)$
7	1.3×10^{11}	$3 : 3393,\ 5 : 3286,\ 6 : 13119$
	1.3×10^{13}	$3 : -10947,\ 5 : 47864,\ 6 : 26129$
	3×10^{14}	$3 : 311978,\ 5 : 310735,\ 6 : 324843$
8	1.3×10^{11}	$3 : 9199,\ 5 : 16125,\ 7 : 8937$
	1.3×10^{13}	$3 : 102728,\ 5 : 126743,\ 7 : 164951$
	3×10^{14}	$3 : 635892,\ 5 : 732174,\ 7 : 505845$
11	1.3×10^{11}	$2 : 2389,\ 6 : 4161,\ 7 : 2134,\ 8 : -2400,\ 10 : 2799$
	1.3×10^{13}	$2 : 5327,\ 6 : 30403,\ 7 : 7351,\ 8 : 74838,\ 10 : 11503$
	3×10^{14}	$2 : 122407,\ 6 : 265991,\ 7 : 334065,\ 8 : 70212,\ 10 : 137533$
19	1.3×10^{11}	$2 : 4934,\ 3 : 8419,\ 8 : 137,\ 10 : 5156,\ 12 : 2974,\ 13 : 5167,\ 14 : 5172,\ 15 : -1918,\ 18 : 9401$
	1.3×10^{13}	$2 : 17964,\ 3 : 60702,\ 8 : 13926,\ 10 : 79470,\ 12 : 30889,\ 13 : 24559,\ 14 : 48327,\ 15 : -5154,\ 18 : 54192$
	3×10^{14}	$2 : 34332,\ 3 : 294945,\ 8 : 103206,\ 10 : 210171,\ 12 : -60071,\ 13 : 126960,\ 14 : -171431,\ 15 : -57410,\ 18 : -16802$
23	1.3×10^{11}	$5 : -1905,\ 7 : 4520,\ 10 : -1682,\ 11 : 803,\ 14 : 1002,\ 15 : 3253,\ 17 : 3576,\ 19 : -4922,\ 20 : -556,\ 21 : 5791,\ 22 : -5114$
	1.3×10^{13}	$5 : 16922,\ 7 : 29658,\ 10 : 43160,\ 11 : -6940,\ 14 : -1663,\ 15 : -23007,\ 17 : -13718,\ 19 : 79327,\ 20 : 30864,\ 21 : 54784,\ 22 : 15725$
	3×10^{14}	$5 : 237020,\ 7 : 158368,\ 10 : 155559,\ 11 : 214431,\ 14 : 168616,\ 15 : 205644,\ 17 : 46103,\ 19 : 130729,\ 20 : 77241,\ 21 : 285454,\ 22 : 294472$

Table 2. Top-decade $K = 25$ reconstruction and observed instability.

N	correlation	RMSE	rank changes	leader changes	fraction -1 leads
7	0.965	0.515	17	13	0.358
8	0.931	0.626	23	15	0.358
11	0.939	0.919	30	26	0.377
19	0.971	1.745	17	8	0.415
23	0.826	1.728	39	23	0.132

3. Conditional examples

The special-value combinations for small moduli are useful test cases for Conjecture 1.2, but they are not unconditional rankings of prime counts. For $N = 3$ and 4 the character groups contain one nonprincipal real character, and the selector of Lemma 1.1 reduces the proposed value to the corresponding real logarithm. For $N = 8$ and 12 several real characters contribute. Any displayed ordering in these cases is conditional on the two-parameter limit, the common normalization, and the independently recomputed $B_N(a)$ values. We therefore do not promote the previous small-modulus tables to theorems in this revision.

4. Open analytic obligations

A theorem-level regularized result requires: (i) an unambiguous definition of the statistic on both the prime and spectral sides; (ii) a proved $T(x)$ regime; (iii) a summed off-diagonal estimate uniform over the prime powers up to x; (iv) justified order interchanges; (v) separate control of prime powers, conductor, gamma, and imprimitive factors; (vi) a specified logarithm branch with a reality proof; and (vii) strict inequalities if a universal ordering is claimed. The earlier Fejér-kernel "independent proof" is omitted because its claimed uniform prime-tail bound was not established.

5. Reproducibility, formalization, and declarations

5.1. **Data and code.** The authoritative ordinary-count file has SHA-256 digest

`57957bdb3ce3243272c3d4b8e9ffe7dfb734b759f48b63becf7ae6f924e1caab.`

The versioned computational package contains the zero generator, reconstruction code, complete tabular outputs, transition ledger, figure source, verifier, and SHA-256 manifests. The extended 3×10^{14} curve is one run; the 567 shared baseline cells through 1.3×10^{13} have an independent comparison. A permanent public archive identifier has not yet been assigned and must be added before submission.

5.2. **Lean scope.** Lean 4 formalizes selected finite and algebraic components only. These are the corrected character-selector identity (including the inverse-class witness modulo 7) and four finite statements about the leading quadratic-nonresidue mean. The checked files contain no `sorry` or `admit`. Lean does not formalize Conjecture 1.2, the explicit-formula estimates, the numerical pipeline, DRH, or this manuscript as a whole.

5.3. **Author contributions.** Shin-ya Koyama initiated the regularized prime-bias framework and developed its analytic formulation and intended number-theoretic applications. Saar Shai designed and performed the ordinary prime-count computations through 3×10^{14}, checked the shared baseline, developed the low-zero spectral reconstruction and verification pipeline, analyzed rank transitions and individual modes, identified and independently certified the very low active modulus-19 mode, formalized selected finite character-algebra and quadratic-nonresidue statements in Lean 4, and wrote the numerical and reproducibility sections. Both authors are responsible for the interpretation, the final mathematical claims, and approval of the submitted version.

5.4. **Computational and AI assistance.** Large-language-model tools assisted with manuscript revision, portions of the computational workflow, code review, and proposals for selected Lean 4 proof code. The character-selector certificate returned through the Aristotle interface was downloaded and recompiled against the stated Lean 4 and Mathlib environment. It does not certify the analytic results or the manuscript as a whole. Numerical claims were checked against versioned data, executable verifiers, and artifact manifests. The authors remain responsible for all mathematical statements, computations, code, interpretations, and final text.

References

[1] M. Aoki and S. Koyama, *Chebyshev's bias against splitting and principal primes in global fields*, J. Number Theory **245** (2023), 233–262.

[2] A. Feuerverger and G. Martin, *Biases in the prime number race*, Experiment. Math. **9** (2000), no. 4, 535–570.

[3] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, American Mathematical Society Colloquium Publications, vol. 53, American Mathematical Society, Providence, RI, 2004.

[4] M. Rubinstein and P. Sarnak, *Chebyshev's bias*, Exp. Math. **2** (1994), no. 3, 173–197.

Department of Mechanical Engineering, Toyo University, 2100 Kujirai, Kawagoe-shi, 350-8585 Japan

Email address: `koyama@toyo.jp`

Independent researcher

Email address: `saar.shai@gmail.com`